Importing files

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Biomedical databases and files

One important step

- Fill a database: In Excel
- Quickly understand formats
- Import the database to R

Excel files

The most frequent software to start a database is MS Excel. It is a versatile program, available in OneDrive, Office 365, MS Office online, and older versions of MS Office. You can also work with Excel in your mobile phone.

Two types of Excel files: "my_database.xlsx" (most well known) and "my_database.xls" (older versions).

Recommendations:

- Do not include text annotations, or use a different column ("Comments").
- Categorize as much as possible.
- Decimals: always use the same format ("," or "."). Never mix.
- Dates: always use the same format.
- Empty fields or missing values: do not write anything
- Sheet1: database itself.
- Sheet2: legend (make it reproducible).

Other files

Sometimes you do not have a software to open a database or the database has a different format (".txt", ".csv", ".tsv"). All of them are human readable.

```
# A text file is a simple plain text like this.
# My name is David

# CSV: comma separated values (or ";")
# Name,NHC,Date_diagnosis,Age,CD22,CD19,Kappa,Lambda,Date_last_follow_up
# David,99X120,24/11/2000,35,2.3,99.9,0.5,0.5,25/12/2005
# Laura,11LP20,11/12/2001,30,0.5,11.4,0.8,0.2,30/01/2010
# Joan,00XYK88,09/10/2010,31,0.6,40.5,0.6,0.4,01/01/2020

# TSV: tab separated values
# Name  NHC Date_diagnosis  Age CD22  CD19  Kappa Lambda  Date_last_follow_up
# David 99X120  24/11/2000  35  2.3 99.9  0.5 0.5 25/12/2005
# Laura 11LP20  11/12/2001  30  0.5 11.4  0.8  0.2  30/01/2010
# Joan  00XYK88 09/10/2010  31  0.6 40.5  0.6 0.4 01/01/2020
```

Import files

You can import any document in R. Some issues can arise when there is no program available to import a specific file.

R works with programs or "packages". Some of them are native (they are included in your base R installation), and some others are dependent on packages that you need to install.

```
# Import excel files: I recommend to install the package "readxl"
# install.packages("readxl")
```

Once the package is installed, you only need to run

```
library(readxl)
```

Then, you can import your excel file and give it the name you want

```
myeloma <- read_excel("/Users/dfmoreno/Dropbox/Lectures/myeloma.xlsx")
```

And you can start exploring the database (dataframe or tibble)

```r
# Check the first lines
head(myeloma)
```

```
# A tibble: 6 x 12
  Patient  molecular_group chr1q21_status treatment event  time   CCND1 CRIM1
  <chr>    <chr>           <chr>          <chr>     <dbl> <dbl>   <dbl> <dbl>
1 GSM50986 Cyclin D-1      3 copies       TT2           0  69.2  9908. 421.
2 GSM50988 Cyclin D-2      2 copies       TT2           0  66.4 16699.  52
3 GSM50989 MMSET           2 copies       TT2           0  66.5   294. 618.
4 GSM50990 MMSET           3 copies       TT2           1  42.7   242.  11.9
5 GSM50991 MAF             <NA>           TT2           0  65     473.  38.8
6 GSM50992 Hyperdiploid    2 copies       TT2           0  65.2   664.  16.9
# i 4 more variables: DEPDC1 <dbl>, IRF4 <dbl>, TP53 <dbl>, WHSC1 <dbl>
```

```r
# Check the last lines
tail(myeloma)
```

```
# A tibble: 6 x 12
  Patient  molecular_group  chr1q21_status treatment event  time   CCND1 CRIM1
  <chr>    <chr>            <chr>          <chr>     <dbl> <dbl>   <dbl> <dbl>
1 GSM51330 Cyclin D-2       2 copies       TT2           0  22.6 17445.   33.6
2 GSM51331 Proliferation    2 copies       TT2           1  17.9   485.   65.7
3 GSM51333 Cyclin D-2       2 copies       TT2           0  22.0 26079    56.8
4 GSM51334 Hyperdiploid     <NA>           TT2           0  21.8  1112.   26.5
5 GSM51335 MMSET            4+ copies      TT2           0  23.8    89.9 165.
6 GSM51336 Low bone disease 2 copies       TT2           1  12.6    55.9 554.
# i 4 more variables: DEPDC1 <dbl>, IRF4 <dbl>, TP53 <dbl>, WHSC1 <dbl>
```

```r
# See the "structure" of the database.
#Gives you all the columns and what "class" or type of variables they contain:
# chr --> character
# num --> numeric (decimals)
# int --> integer (no decimals)
# factor --> factor (positive OR negative)
str(myeloma)
```

```
tibble [256 x 12] (S3: tbl_df/tbl/data.frame)
 $ Patient        : chr [1:256] "GSM50986" "GSM50988" "GSM50989" "GSM50990" ...
 $ molecular_group: chr [1:256] "Cyclin D-1" "Cyclin D-2" "MMSET" "MMSET" ...
 $ chr1q21_status : chr [1:256] "3 copies" "2 copies" "2 copies" "3 copies" ...
```

```
 $ treatment      : chr [1:256] "TT2" "TT2" "TT2" "TT2" ...
 $ event          : num [1:256] 0 0 0 1 0 0 1 0 1 1 ...
 $ time           : num [1:256] 69.2 66.4 66.5 42.7 65 ...
 $ CCND1          : num [1:256] 9908 16699 294 242 473 ...
 $ CRIM1          : num [1:256] 420.9 52 617.9 11.9 38.8 ...
 $ DEPDC1         : num [1:256] 523.5 21.1 192.9 184.7 212 ...
 $ IRF4           : num [1:256] 16156 16946 8904 11895 7563 ...
 $ TP53           : num [1:256] 10 1057 1763 947 361 ...
 $ WHSC1          : num [1:256] 262 364 10043 4931 165 ...
```

You can also import csv and tsv files

```r
# It is good practice to include the separator (sep)
# read.csv can read multiple formats, not only csv files
myeloma_commas <- read.csv("/Users/dfmoreno/Dropbox/Lectures/myeloma_commas.csv", sep=",")
myeloma_tabs <- read.csv("/Users/dfmoreno/Dropbox/Lectures/myeloma_tabs.tsv", sep="\t")

# Explore the dataframe
head(myeloma_commas)
```

```
  X  Patient molecular_group chr1q21_status treatment event  time   CCND1 CRIM1
1 1 GSM50986      Cyclin D-1       3 copies       TT2     0 69.24  9908.4 420.9
2 2 GSM50988      Cyclin D-2       2 copies       TT2     0 66.43 16698.8  52.0
3 3 GSM50989           MMSET       2 copies       TT2     0 66.50   294.5 617.9
4 4 GSM50990           MMSET       3 copies       TT2     1 42.67   241.9  11.9
5 5 GSM50991             MAF           <NA>       TT2     0 65.00   472.6  38.8
6 6 GSM50992    Hyperdiploid       2 copies       TT2     0 65.20   664.1  16.9
  DEPDC1    IRF4   TP53   WHSC1
1  523.5 16156.5   10.0   261.9
2   21.1 16946.2 1056.9   363.8
3  192.9  8903.9 1762.8 10042.9
4  184.7 11894.7  946.8  4931.0
5  212.0  7563.1  361.4   165.0
6  341.6 16023.4 2096.3   569.2
```

```r
head(myeloma_tabs)
```

```
   Patient molecular_group chr1q21_status treatment event  time   CCND1 CRIM1
1 GSM50986      Cyclin D-1       3 copies       TT2     0 69.24  9908.4 420.9
2 GSM50988      Cyclin D-2       2 copies       TT2     0 66.43 16698.8  52.0
3 GSM50989           MMSET       2 copies       TT2     0 66.50   294.5 617.9
```

```
4 GSM50990          MMSET     3 copies      TT2      1 42.67   241.9  11.9
5 GSM50991            MAF         <NA>      TT2      0 65.00   472.6  38.8
6 GSM50992    Hyperdiploid    2 copies      TT2      0 65.20   664.1  16.9
  DEPDC1    IRF4   TP53   WHSC1
1  523.5 16156.5   10.0   261.9
2   21.1 16946.2 1056.9   363.8
3  192.9  8903.9 1762.8 10042.9
4  184.7 11894.7  946.8  4931.0
5  212.0  7563.1  361.4   165.0
6  341.6 16023.4 2096.3   569.2
```